

## AI/ML Internship – Day 36 Notes & Tasks

### Module 7: Train-Test Split & Model Evaluation

---

#### Part 1: Why Model Evaluation is Important?

 In Machine Learning:

- Training alone is not enough
- We must check whether model performs correctly on new data

 This process is called:

 Model Evaluation

---

#### Part 2: What is Train-Test Split?

##### Definition

 Train-Test Split means dividing dataset into:

- Training Data
  - Testing Data
- 

#### Part 3: Why Split the Dataset?

 If model sees all data during training:

- It may memorize data
- Predictions become unrealistic

 So:

- Train data → used for learning
  - Test data → used for evaluation
- 

#### Part 4: Theory Answers (IMPORTANT)

---

##### 1. What is Train Data?

👉 Data used to train the ML model.

---

## ◆ 2. What is Test Data?

👉 Data used to check model performance.

---

## ◆ 3. What is Train-Test Split?

👉 Process of dividing dataset into training and testing parts.

---

## ◆ 4. Why Model Evaluation is Important?

👉 To measure:

- Accuracy
  - Performance
  - Prediction quality
- 

## ◆ 5. What is Accuracy?

👉 Accuracy shows how correct the model predictions are.

---

## 📌 Part 5: Import Libraries

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import LinearRegression
```

---

## 📌 Part 6: Create Dataset

```
data = {  
    "Hours": [1, 2, 3, 4, 5, 6, 7, 8],  
    "Marks": [20, 35, 50, 65, 80, 85, 90, 95]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

---

### **Part 7: Features & Labels**

```
X = df[["Hours"]]
```

```
y = df["Marks"]
```

---

👉 X = input/features

👉 y = output/labels

---

### **Part 8: Splitting Dataset**

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

---

### **Part 9: Understanding Parameters**

| Parameter       | Meaning              |
|-----------------|----------------------|
| test_size=0.2   | 20% data for testing |
| random_state=42 | Fixed random split   |

---

### **Part 10: Train the Model**

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

---

👉 Model learns only from training data.

---

### ✦ Part 11: Make Predictions

```
predictions = model.predict(X_test)
```

```
print(predictions)
```

---

### ✦ Part 12: Compare Actual vs Predicted

```
result = pd.DataFrame({  
    "Actual": y_test,  
    "Predicted": predictions  
})
```

```
print(result)
```

---

### ✦ Part 13: Model Accuracy Score

```
score = model.score(X_test, y_test)
```

```
print("Accuracy:", score)
```

---

👉 score() gives model performance.

---

### ✦ Part 14: Complete ML Workflow

Dataset



Train-Test Split



Train Model



Test Model



Prediction

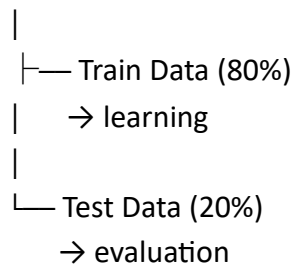


Evaluation

---

## 📌 Part 15: Visualization of Train-Test Split

Total Dataset



---

## 📌 Part 16: Real-World Thinking

👉 Real AI systems always use testing.

Examples:

- Face recognition
- Spam detection
- Medical prediction
- Recommendation systems

👉 Without testing:

❌ Model reliability cannot be trusted.

---

## 📌 Part 17: Overfitting Concept (Basic Introduction)

### ◆ What is Overfitting?

👉 When model memorizes training data too much.

Result:

- Performs well on training data
- Performs poorly on new data

👉 Train-test split helps reduce this problem.

---

## 📌 Part 18: Why `random_state` is Important?

👉 Ensures:

- Same split every time
  - Reproducible results
- 

## Tasks for Day 36

---

### Task 1: Theory (Write Answers)

1. What is train data?
  2. What is test data?
  3. What is train-test split?
  4. Why model evaluation is important?
  5. What is accuracy?
- 

### Task 2: Dataset Practice

1. Create student dataset
  2. Create salary dataset
  3. Create sales dataset
- 

### Task 3: Train-Test Split Practice

1. Split dataset
  2. Use `test_size`
  3. Use `random_state`
- 

### Task 4: Model Training

1. Train linear regression model
  2. Use `fit()`
  3. Train with training data only
-

### **Task 5: Prediction Practice**

1. Predict test data
  2. Compare actual vs predicted
  3. Print accuracy score
- 

### **Task 6: Result Analysis**

1. Identify good predictions
2. Identify prediction errors
3. Explain accuracy meaning